# Benedetta Mariani


Born: on 21 December 1995 | Nationality: Italian

Department of Physics, University of Padua (Via Marzolo 8 Padua), Padova Neuroscience Center (Via Orus 2 Padua)

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## RESEARCH EXPERIENCES

*October 2024 - present day*

### Post-doctoral researcher

Padova Neuroscience Center
Supervisors: prof. Michele Allegra and prof. Samir Suweis
Inference of directed interactions in the brain through fMRI data.
Founded by the PNC Project INSIGHTS (PNRR Cascading Grant) within the Research and Innovation Program titled "MNESYS" (an integrated multiscale approach to the study of the nervous system in health and disease).

*October 2020 - September 2024*

### PhD student in Neuroscience

Padova Neuroscience Center
Laboratory of Intedisciplinary Physics, Padova https://www.liphlab.com/

- Supervisor: Prof Samir Suweis
- Main research interests: combining the tools of physics of complex systems and data analysis to study collective behavior in Neuroscience and Systems biology (e.g. how computation in neural systems is shaped by the dynamical regime and/or by oscillatory behaviors)

*December 2019 – September 2020*

### Research scholarship

Laboratory of interdisciplinary Physics, University of Padua

- Project title: "Criticality in rat somatosensory barrel cortex"
- Supervisors: Prof Samir Suweis and Prof. Stefano Vassanelli.

## EDUCATION

*December 2017 – October 2019*

### Master degree in Physics of Complex Systems

University of Turin, Italy

- Final grade: 110/110 cum laude
- Final project: "Signatures of criticality in rat brain: beyond avalanches statistics" permormed at Liph Lab (Physics and Astronomy Dept, Padua) and at NeuroChip Lab (Dept. of Biomedical Sciences, Padua).

*September 2015 – December 2017*

### Bachelor degree in Physics

University of Padua, Italy

- Final grade: 100/110
- Final project: "Phase separation for intrinsically disordered proteins"

*October 2014 – September 2015*

### Student of Biomedical Engineering

Polytechnic University of Milan

## PUBLICATIONS

November 2023

### Prenatal experience with language shapes the brain

- Science Advances Vol 9, Issue 47
- Benedetta Mariani, Giorgio Nicoletti, Giacomo Barzon, Maria Clemencia Ortiz Barajas, Mohinish Shukla, Ramon Guevara, Samir Simon Suweis, Judit Gervain
- https://www.science.org/doi/10.1126/sciadv.adj3524; https://www.aaas.org/news/babies-brains-are-primed-their-native-language-birth

July 2022 **Criticality and network structure drive emergent oscillations in a stochastic whole-brain model**

- Journal of Physics: Complexity, Volume 3, Number 2
- Giacomo Barzon, Giorgio Nicoletti, Benedetta Mariani, Marco Formentin, Samir Suweis
- https://iopscience.iop.org/article/10.1088/2632-072X/ac7a83/meta

June 2022 **Disentangling the critical signatures of neural activity**

- Scientific Reports volume 12, Article number: 10770
- Benedetta Mariani, Giorgio Nicoletti, Marta Bisio, Marta Maschietto, Stefano Vassanelli, Samir Suweis
- https://www.nature.com/articles/s41598-022-13686-0

August 2021 **Neuronal Avalanches Across the Rat Somatosensory Barrel Cortex and the Effect of Single Whisker Stimulation**

Front. Syst. Neurosci.,vol 15
Benedetta Mariani, Giorgio Nicoletti, Marta Bisio, Marta Maschietto, Roberto Oboe, Alessandro Leparulo, Samir Suweis, Stefano Vassanelli

- https://www.frontiersin.org/articles/10.3389/fnsys.2021.709677/full

## PREPRINTS

July 2025 **The somatosensory barrel cortex controls the spindle thalamocortical oscillation by frequency locking**

- bioRxiv 2025.07. 09.662963
- Mattia Tambaro, Ramón Guevara, Marta Maschietto, Alessandro Leparulo, Claudia Cecchetto, Giorgio Nicoletti, Benedetta Mariani, Samir Suweis, Stefano Vassanelli

June 2025 **Whisker stimulation reinforces a resting-state network in the barrel cortex: nested oscillations and avalanches**

- bioRxiv 2025.06.06.658305
- Benedetta Mariani, Ramón Guevara, Mattia Tambaro, Marta Maschietto, Alessandro Leparulo, Stefano Vassanelli, Samir Suweis

## TEACHING EXPERIENCE

*October 2025* **Teaching assistant**

- for the "Advanced Machine learning" module taught by prof. Nicolò Navarin at the Master in "Quantitative and Computational Biosciences" (UniPd)
- Topics: training of deep feedforward neural network, convolutional network, recurrent neural network, graph convolutional networks, generative models

## THESIS CO-SUPERVISION

*December 2022* Co-supervision of a UniPd Physics bachelor degree thesis: "Analyses of neural oscillations and synchrony in the brain"

## ORGANIZATION OF CONFERENCES

*September 2025* Part of the Local Organizing Committee of INCTN (Italian Network for Computational and Theoretical Neuroscience) 2025, Padua

## CONFERENCE ABSTRACTS

*June 2026 (upcoming)* **OHBM 2026, Bordeaux, France**

Towards consistent and scalable whole-brain effective connectivity estimation
Benedetta Mariani, Luca Taffarello, Giacomo Barzon, Michele Allegra, Stefano de Marchi, Antonino Vallesi, Marco Zorzi, Samir Suweis, Alessandra Bertoldo

*March 2026* **Cosyne 2026, Lisbon, Portugal**

Self-sustained oscillations link spontaneous and stimulus-driven dynamics in the barrel cortex
Benedetta Mariani, Ramon Guevara, Mattia Tambaro, Giorgio Nicoletti, Marta Maschietto, Alessandro Leparulo, Stefano Vassanelli, Samir Suweis

*October 2025*
### Bernstein Conference 2025, Frankfurt am Main
Nested oscillations and critical avalanches prime the rat barrel cortex for whisker stimuli
Samir Suweis, Benedetta Mariani, Ramon Erra Guevara, Mattia Tambaro, Marta Maschietto, Alessandro Leparulo, Stefano Vassanelli Presenter: Samir Suweis

*July 2025*
### Annual Computational Neuroscience Meeting (OCNS) 2025, Florence
Spontaneous oscillations and neural avalanches are linked to whisker stimulation response in the rat-barrel and thalamus circuit
Benedetta Mariani, , Ramon Guevara Erra, Mattia Tambaro, Marta Maschietto , Alessandro Leparulo, Stefano Vassanelli, , and Samir Suweis

*September 2022*
### Bernstein Conference 2022, Berlin
Disentangling the critical signatures of neural activity: avalanches, spatial correlations and information.
Giorgio Nicoletti, Benedetta Mariani, Marta Bisio, Marta Maschietto, Stefano Vassanelli, Samir Suweis. Presenter: Giorgio Nicoletti

*September 2022*
### Bernstein Conference 2022, Berlin
Collective oscillations in the rat barrel-thalamus network
Benedetta Mariani, Ramon Guevara Erra, Giorgio Nicoletti, Stefano Vassanelli, Samir Suweis https://abstracts.g-node.org/conference/BC22/abstracts#/uuid/83d6de84-d449-4f47-ba63-7cd486e93e40

September 2021
### Bernstein Conference 2021, online
Disentangling the role of external and intrinsic dynamics on the critical signatures of neural activity https://abstracts.g-node.org/conference/BC21/abstracts#/uuid/00a75410-d18a-4ba6-a14d-aa682e2c8777

*October 2020*
### Bernstein Conference 2020, online
Brain criticality beyond resting state neuronal avalanches https://abstracts.g-node.org/abstracts/b9ec73de-f9db-4df2-9ab1-c08c2c532e65

## MAIN TALKS

*June 2025*
### Contributed talk at the ”Collective Dynamics and Information Processing in Neural Systems” StatPhys29 Satellite, Venice
Whisker stimulation reinforces a resting-state network in the barrel cortex: nested oscillations and avalanches

*February 2025*
### Invited talk at the SIFS Young series seminar
Spontaneous oscillations and avalanches are linked to whisker sensory response processing in the rat thalamocortical circuit

*May 2024*
### Invited Talk, ”Baby Rhythm” ERC closing workshop, Palazzo Bo, Padova, Italy
”Prenatal experience with language shapes the brain”

*October 2020*
### Contributed talk at Brain Criticality Virtual Meeting, online
”Brain criticality beyond resting state neuronal avalanches” https://braincriticality.org/2020/10/11/benedetta-mariani-brain-criticality-beyond-resting-state-neuronal-avalanches-spotlight-talk/

## OTHER PRESENTATIONS AND POSTERS

*September 2025*
### Poster Presentation at INCTN 2025, Padua
- Whisker stimulation reinforces a resting-state network in the barrel cortex: nested oscillations and avalanches

*September 2024*
### Poster presentation at Italian Network for computational Neuroscience, Rome
- Spontaneous 10-15 Hz oscillations in the rat barrel cortex stabilize after whisker stimulation and are modulated by slow thalamic oscillations

*February 2023* **Talk at Liph Lab Spring Workshop, Asiago (VI), Italy**
- ”Collective oscillations in the rat barrel-thalamus network”

*31st March 2022* **Talk at Liph Lab Spring Workshop, Asiago (VI), Italy**
- ”Modelling oscillations in the rat barrel-thalamus network”

*November 2022* **Poster at Brain Criticality Hybrid Meeting November 7-9 2022, Online**
- ”Collective oscillations in the rat barrel-thalamus network”
- https://braincriticality.org/2022/11/07/poster-20227-benedetta-mariani-collective-oscillations-in-the-rat-barrel-thalamus-network/

*June 2021* **Poster at the Beg Rohu Summer School, St. Pierre Quiberon, France**
- ”On the critical signatures of neural activity”

*January 2020* **Talk at Liph Lab winter Workshop, Folgaria (TN)**
- I presented my research outputs.

## ATTENDED CONFERENCES AND SCHOOLS

*28-31 May 2024* **Stochastic Models and Experiments in Ecology and Biology, Gran Sasso Science Institute, L’Aquila, Italy**

*May 2024* **”Baby Rhythm”, Judit Gervain ERC closing workshop, Palazzo Bo, Padova**

*November 2022* **Brain Criticality Hybrid Meeting November 7-9 2022, Online**

*September 2022* **Bernstein Conference 2022, Berlin**

*16th-20th March 2022* **Cosyne Tutorial and Cosyne Main meeting, Lisbon**

*15th-24th September 2021* **EITN Fall School in Computational Neuroscience, Paris**
- I took part in the project group ”Mean field and population models” and presented the outputs of our project.
- I attended lectures and tutorials on different models and sofwares to simulate networks of neurons at different levels of complexity (Brian, Nest, Neuron, TVB)

*September 2021* **Bernstein Conference 2021, online**

*31st May - 12nd June 2021* **The Beg Rohu Summer School, St. Pierre Quiberon, France**

*23rd February - 26th February 2021* **Computational and Systems Neuroscience (Cosyne) 2021, online**

*14th - 16th December 2020* **Brain Hack Padova, online**
- I led the project group ”Association of Microstructural white matter and personality traits based on Human connectome project dataset”

*6th October - 9th October 2020* **Brain Criticality Virtual Meeting, online**

*29th September - 1st October 2020* **Bernstein Conference 2020, online**

*18th - 23th July 2020* **29th Annual Computational Neuroscience Meeting, Organization for Computational Neuroscience, online**

*29th June - 3rd July 2020* **Youth in High-dimensions: Machine Learning, High-dimensional Statistics and Inference for the New Generation, online**

*9th - 14th September 2019* Summer School in Computational and Theoretical Models in Neuroscience, European centre for living technology, Venice

## AWARDS

*November 2023* We published "Prenatal experience with language shapes the brain" https://www.science.org/doi/10.1126/sciadv.adj3524?utm_campaign=SciMag&utm_source=Twitter&utm_medium=ownedSocialin the high impact journal "Science Advances" https://www.aaas.org/news/babies-brains-are-primed-their-native-language-birth and gained visibility (journalists from many countries quoted and interviewed us, including "Le Scienze" journal, "Nature Italy" journal, "El Pais, Spain", "The Times, UK", "Le Figaro, France", "New Scientist, UK", e.g.: https://www.lescienze.it/news/2023/11/23/news/linguaggio_mamma_cervello_neonati-14259373/; https://www.nature.com/articles/d43978-023-00181-x; and we participated to podcasts broadcasting our work, e.g: https://www.thenakedscientists.com/articles/interviews/babies-learn-language-they-are-even-born

*2022* Our paper "Disentangling the critical signatures of neural activity" https://www.nature.com/articles/s41598-022-13686-0 has been included in the Top 100 Neuroscience paper of 2022 Scientific Reports Journal.

*2021* Winner of SECS (Scholarships for Events on Complex Systems) grant, provided by the Young Researchers of the Complex Systems Society.

*2014* Winner of the award for the high school students with the highest final grade and honors, granted by MIUR, the Italian Ministry for Education, University and Research.

## COMPUTER SKILLS

Programming Languages

- Python: advanced level
- C++, Mathematica, R, matlab: intermediate level

Operating systems

- Windows: advanced level
- Unix: advanced level

## PERSONAL SKILLS

Languages known

- Mother tongue: Italian
- English: level B2 (First Certificate of English)
- German: level A2 (certificate from the Goethe Institute of Bremen)